

5G RAN

Distributed Antenna Solutions (Low-Frequency TDD) Feature Parameter Description

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Date	2025-12-31



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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Revised the constraints affecting cell activation due to license insufficiency. For details, see 4.3.1 Licenses .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added support for CSI measurement optimization in distributed massive MIMO cells. For details, see 4.2.1 Benefits .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
 - Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-050202	Distributed Massive MIMO	4 Distributed Massive MIMO
FOFD-071211	Distributed Massive MIMO (NR TDD)	

 NOTE

FOFD-050202 Distributed Massive MIMO is a feature for LampSite base stations, and
FOFD-071211 Distributed Massive MIMO (NR TDD) is a feature for macro base stations.

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Distributed massive MIMO	Uplink beam boosting for distributed massive MIMO is supported only in SA networking. For other functions, there is no difference.	4 Distributed Massive MIMO

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Distributed massive MIMO	Supported only in low frequency bands	4 Distributed Massive MIMO

3 Overview

Distributed antennas can be deployed to form a distributed massive MIMO cell. Antennas of multiple transmission reception points (TRPs) are used at transmit and receive ends to implement communication in conjunction with signal processing techniques. This reduces interference and increases UE throughput in overlapping areas.

4 Distributed Massive MIMO

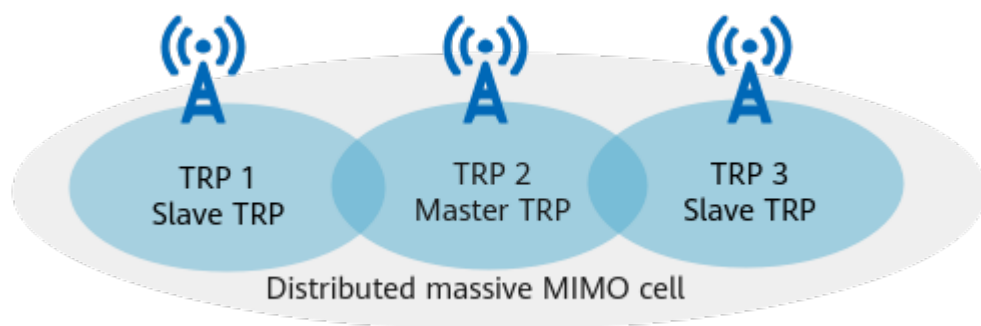
4.1 Principles

Cell split can be used for capacity expansion in high-capacity scenarios. After the split, however, the capacity may not increase or even decrease due to severe intra-NR-frequency inter-cell interference. In addition, deep fading is likely to occur in indoor coverage scenarios, affecting user performance. Distributed massive MIMO is introduced to address these issues. The basic functions of distributed massive MIMO can be enabled by selecting the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter.

A common cell is served by only one TRP, whereas a distributed massive MIMO cell is served by multiple TRPs, as shown in Figure 4-1. Among the TRPs of a distributed massive MIMO cell, one is the master TRP and the others are slave TRPs. Only the master TRP needs to be assigned baseband equipment while the slave TRPs are configured with only RF information. The master and slave TRPs share the same baseband equipment. The master TRP is specified by setting the **NRDUCellTrp.TrpType** parameter to **MASTER_DM_MIMO**. A slave TRP is specified by setting the **NRDUCellTrp.TrpType** parameter to **SLAVE** and associated with the master TRP using the **NRDUCellTrp.MasterTrpId** parameter. (A maximum of 12 slave TRPs can be configured.) The more the TRPs serving a distributed massive MIMO cell, the larger the decrease in the number of handovers.

- For LampSite base stations, distributed massive MIMO eliminates cell edges by combining n 4T4R/2T2R cells that provide contiguous coverage and are served by RF modules working on the same frequency band into one $4n$ T $4n$ R cell. In this scenario, only 4T4R TRPs or both 4T4R and 2T2R TRPs can be combined to serve a distributed massive MIMO cell, but 2T2R TRPs cannot be combined to do so.
- For ToB services of TDD macro base stations, distributed massive MIMO eliminates cell edges by combining n 4T4R/8T8R cells that provide contiguous coverage and are served by RF modules working on the same frequency band into one $4n$ T $4n$ R or $8n$ T $8n$ R cell. In this scenario, either 4T4R or 8T8R TRPs can be combined to serve a distributed massive MIMO cell, but 4T4R and 8T8R TRPs cannot be used together.

Figure 4-1 Distributed massive MIMO cell



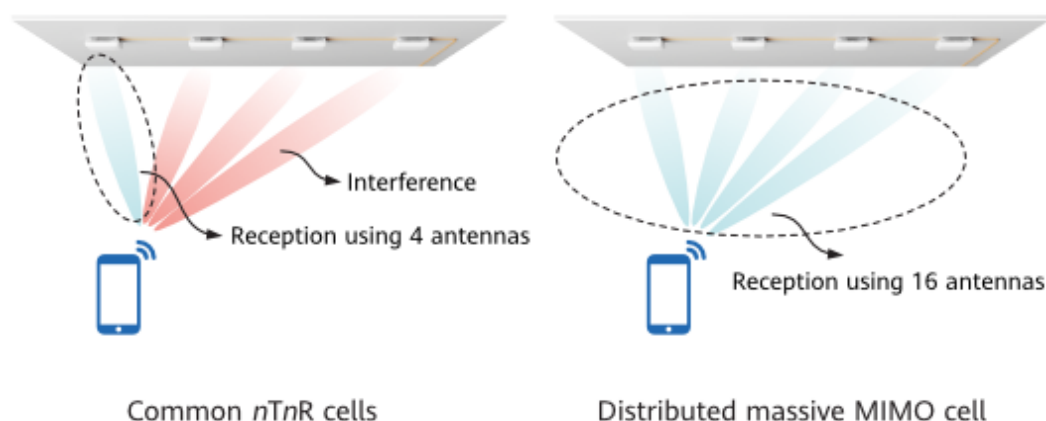
The following sections describe the key techniques of distributed massive MIMO. This function uses multi-antenna joint transmission/reception and joint scheduling to improve single-UE performance (see [Multi-antenna Joint Transmission/Reception and Joint Scheduling](#)), and uses multi-user resource multiplexing to improve cell capacity (see [Multi-user Resource Multiplexing](#)). In addition, distributed massive MIMO provides enhancement functions for special scenarios (see [Enhancement Functions](#)).

Multi-antenna Joint Transmission/Reception and Joint Scheduling

Distributed massive MIMO uses multi-antenna joint transmission/reception and joint scheduling to decrease the number of handovers for UEs moving between TRPs in a cell and improve the performance of UEs at the coverage edges of the TRPs. In each common $nTnR$ cell, UEs exchange their uplink and downlink data only with the antennas of their respective serving cells. Scheduling is also separately performed in these cells. UEs at the coverage edges of the TRPs are vulnerable to interference from neighboring cells. In a distributed massive MIMO cell, there are more transmit and receive antennas, which improves the uplink and downlink user experience and reduces inter-cell interference.

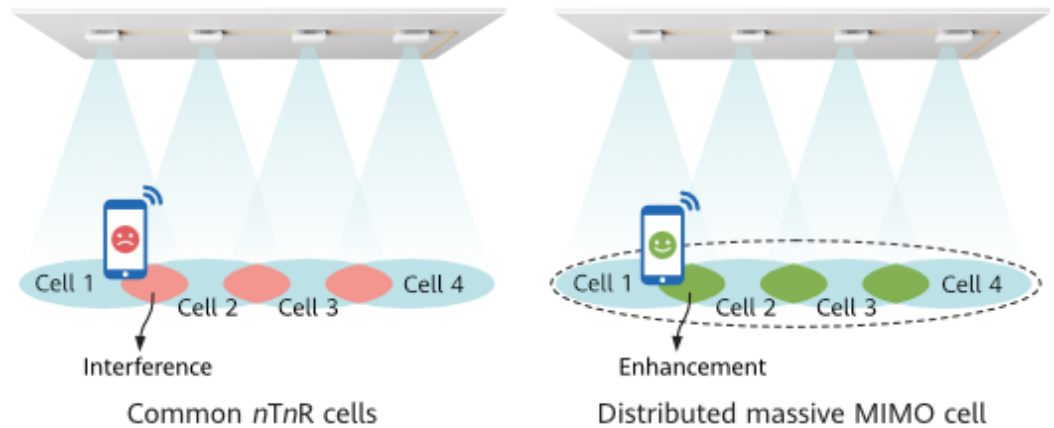
- In the uplink, the distributed TRPs of the distributed massive MIMO cell jointly receive signals, which improves the received signal strength and anti-interference capabilities, as shown in [Figure 4-2](#).

Figure 4-2 Example of distributed massive MIMO in the uplink (using the combination of four 4T4R TRPs)



- In the downlink, the distributed massive MIMO cell performs joint weighting based on the distributed TRPs, for better matching with radio channel characteristics, which improves the downlink signal strength for UEs, as shown in [Figure 4-3](#).

Figure 4-3 Example of distributed massive MIMO in the downlink



Multi-user Resource Multiplexing

Distributed massive MIMO uses MU-MIMO to allow for spatial multiplexing of PDSCH, PDCCH, and PUSCH resources for multiple UEs, which boosts spectral efficiency, as shown in [Figure 4-4](#). This results in significantly higher cell capacity, cell throughput, and UE throughput than that under the single-TRP solution. A distributed massive MIMO cell sends periodic CSI-RSs in joint transmission mode. The gNodeB determines whether UEs are in overlapping areas based on the SRS RSRP differences between TRPs. For UEs in overlapping areas, multiple TRPs jointly process traffic channel data to increase the UE throughput. For UEs not in overlapping areas, the beams of multiple TRPs can use the same time-frequency resources of PUSCH, PDSCH, or PDCCH for spatial multiplexing. For the maximum number of spatial multiplexing layers for multiple UEs in distributed massive MIMO, see [Table 4-1](#). The `NRDUCellPdcchAlgo.PdcchUserNumberThldCoeff` parameter can be used to specify the coefficient of the UE number threshold for PDCCH symbol number adaptation.

Figure 4-4 MU-MIMO

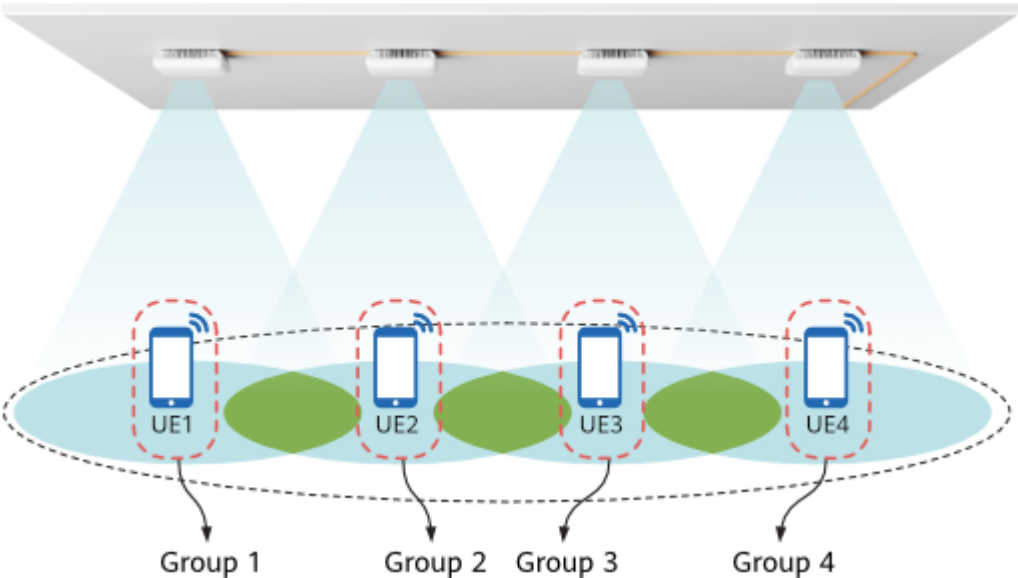


Table 4-1 Maximum number of spatial multiplexing layers for multiple UEs in distributed massive MIMO

Number of Antennas After the Combination	Maximum Number of PDSCH MU-MIMO Layers	Maximum Number of PDCCH MU-MIMO Layers	Maximum Number of PUSCH MU-MIMO Layers
16T16R or more	16	4	8
8T8R	8	Not supported	4
The maximum number of spatial multiplexing layers allowed in a cell is limited by the number of layers supported by the baseband resources of the cell.			

Enhancement Functions

- For LampSite base stations, the following enhancement functions are provided based on the scenario requirements:
 - In heavy-load scenarios with insufficient PDCCH resources, PDCCH beam optimization for distributed massive MIMO can be enabled to increase the probability of PDCCH spatial multiplexing. This function is controlled by the **DM_MIMO_PDCCH_BEAM_OPT_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter.
 - In heavy-load scenarios with insufficient PDCCH resources, if a LampSite cell experiences severe interference from a neighboring macro cell, TRP-level PDCCH transmission for distributed massive MIMO can be enabled by selecting the **DM_MIMO_PDCCH_TRP_TRANS_SW** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter. This function requires that the distributed massive MIMO cell be only served by low-frequency TDD LampSite 4T4R TRPs, jointly served by low-frequency TDD LampSite

- 4T4R and 2T2R TRPs, or jointly served by low-frequency TDD LampSite 4T4R TRPs and DAS 2T RRUs.
- In a distributed massive MIMO cell, TRP-based weight selection for distributed massive MIMO can be enabled to select different TRPs and weights for different UEs based on the UE location (edge or center). This function is controlled by the **DM_MIMO_TRP_SEL_WEIGHT_SW** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter. After this function is enabled, appropriate TRPs are selected for downlink data transmission based on the SRS RSRP measured by each TRP, improving downlink service experience.
 - For macro base stations with 8T8R TRPs in ToB scenarios, uplink beam boosting for distributed massive MIMO can be enabled by selecting the **DM_MIMO_UL_BEAM_BOOST_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter. After this function is enabled, the base station optimizes the directions of uplink receive beams and fully utilizes the uplink beamforming capability of the antenna array, increasing the RSRP of the optimal beam and improving the uplink capacity.

4.2 Network Analysis

4.2.1 Benefits

This function reduces inter-cell interference in overlapping areas and increases UE throughput by increasing SINRs in overlapping areas. The uplink and downlink gains may vary with configurations.

Inter-RRU channel calibration for distributed massive MIMO can be enabled in a distributed massive MIMO cell to further improve the weight accuracy and downlink service experience.

- For LampSite base stations, this function is controlled by the **INTER_RRU_CHN_CALIB_SW** option of the **NRDUCellAlgoSwitch.ChnCalibSwitch** parameter.
- For macro base stations, this function is controlled by the **DM_MIMO_INTER_RRU_CHN_CALIB_SW** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter.

If the aperiodic CSI-RS resources are insufficient for UEs in non-overlapping areas in a distributed massive MIMO cell, the CSI measurement optimization function can be enabled to adaptively adjust periodic CSI-RS resources and thereby prevent CSI measurement failures for these UEs. This function is controlled by the **CSI_MEAS_OPT_SW** option of the **NRDUCellCsirs.CsiRsExtSwitch** parameter. For details about CSI measurement optimization, see *Channel Management*.

Distributed massive MIMO improves system availability for different types of base stations, with the following differences:

- For LampSite base stations, the pRRUs connected to different RHUBs of the same distributed massive MIMO cell provide overlapping coverage. In this case, if an RHUB or pRRU is faulty, UEs can stay connected through adjacent pRRUs without service interruption. In this way, the system availability is improved, but system performance deteriorates.

- For macro base stations, the different RRUs serving the same distributed massive MIMO cell provide overlapping coverage. In this case, if an RRU is faulty, UEs can stay connected through adjacent RRUs without service interruption. In this way, the system availability is improved, but system performance deteriorates.

 NOTE

Distributed massive MIMO uses multiple antennas for joint transmission and reception to reduce inter-UE interference. Theoretically, the maximum number of paired layers is equal to the number of antennas. However, it is recommended that the number of paired layers be less than the number of antennas to obtain more spatial freedom and achieve better interference suppression capability. For example, if 4 TRPs with 16 antennas are configured, the maximum number of layers is 16 in theory. However, the recommended number of layers in a cell is less than 16 (for example, 12) to achieve better inter-UE interference suppression capability. If 6 to 8 TRPs with 24 to 32 antennas are configured, the maximum number of layers in a cell does not need to be limited because it is less than the number of antennas for joint transmission and reception.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

If more than four 4T4R/8T8R cells are combined into one distributed massive MIMO cell, the capacity of the combined cell is less than the total capacity of the original cells.

For LampSite base stations, the transmit power of pRRUs in a distributed massive MIMO cell must be the same as the value of **NRDUCellTrp.MaxTransmitPower** set for the master TRP. Otherwise, the gains will decrease. For details about pRRU power configuration, see *Power Control*.

Enhanced functions for LampSite base stations:

- After PDCCH beam optimization for distributed massive MIMO is enabled in heavy-load scenarios where PDCCH resources are insufficient, the average uplink and downlink cell throughputs increase.
- After TRP-level PDCCH transmission for distributed massive MIMO is enabled in heavy-load scenarios where the LampSite cells experience severe interference from neighboring macro cells and PDCCH resources in the LampSite cells are insufficient, the service drop rate of UEs in the LampSite cells decrease.
- After TRP-based weight selection for distributed massive MIMO is enabled, the average downlink UE throughput in a cell increases in light- or medium-load scenarios and decreases in heavy-load scenarios.

Enhanced functions for macro base stations:

- After uplink beam boosting for distributed massive MIMO is enabled, the base station optimizes the directions of uplink receive beams and fully utilizes the uplink beamforming capability of the antenna array, improving the RSRP of the optimal beam and the uplink capacity.

4.3 Requirements

4.3.1 Licenses

For LampSite base stations, distributed massive MIMO requires the following feature and capacity licenses.

Table 4-2 Feature licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-050202	Distributed Massive MIMO	NR0SDMMIMO00	per Carrier per pRRU
For the license control item NR0SDMMIMO00: - If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. - If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

Table 4-3 Capacity licenses

Model	Description	Sales Unit
NR0S0DLEPU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell
NR0S0ULEPU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell
NR0SBPMMBW00	UBBP Massive MIMO Operating Bandwidth 1*20MHz License (NR)	per 1*20MHz
NR0SDMMIMO03	Distributed Massive MIMO for 5000 Series LampSite pRRU (NR)	per Carrier per pRRU
Note: The maximum number of spatial multiplexing layers cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see <i>License Control Item Lists</i>.		

NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

For macro base stations, distributed massive MIMO requires the following feature and capacity licenses.

Table 4-4 Feature licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
TDD	FOFD-071211	Distributed Massive MIMO (NR TDD)	NR0S00VMMM00	Per Cell
For the license control item NR0S00VMMM00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, both selecting the DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter and setting the NRDUCellTrp.TrpType parameter to MASTER_DM_MIMO is not allowed due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.				

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

Table 4-5 Capacity licenses

Model	Description	Sales Unit
NR0S0DLEPU00	Massive MIMO DL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell
NR0S0ULEPU00	Massive MIMO UL 2-Layers Extended Processing Unit License (NR)	per 2 Layers per Cell
NR0SBPMMBW00	UBBP Massive MIMO Operating Bandwidth 1*20MHz License (NR)	per 1*20MHz

Model	Description	Sales Unit
Note: The maximum number of spatial multiplexing layers cannot exceed the licensed number. One license unit corresponds to two layers. For details about licensing rules, see <i>License Control Item Lists</i> .		

NOTE

The unavailability of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected If the License Is Unavailable** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
PDCCH beam optimization	DM_MIMO_PDCCH_BEAM_OPT_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>MIMO (TDD)</i>	If TRP-level PDCCH transmission is enabled, PDCCH beam optimization must also be enabled.
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	PDCCH beam optimization for distributed massive MIMO, uplink beam boosting for distributed massive MIMO, and TRP-based weight selection for distributed massive MIMO require distributed massive MIMO.

4.3.2.2 Mutually Exclusive Functions

- Basic functions of distributed massive MIMO

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNetworking-Mode set to HYPER_CELL	<i>Hyper Cell</i>	Distributed massive MIMO does not work in hyper cells.

Function Name	Function Switch	Reference	Description
Cell Combination	NRDUCell.NrDuCellNetworking-Mode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Distributed massive MIMO does not work in combined cells.
Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	For macro base stations, distributed massive MIMO does not work with intra-base-station DL CoMP.
Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Distributed massive MIMO does not work with intra-base-station UL CoMP.
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with RF channel intelligent shutdown.
Inter-base-station UL CoMP	INTER_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	None	Distributed massive MIMO does not work with inter-base-station UL CoMP.
Inter-base-station downlink joint transmission	INTER_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	None	Distributed massive MIMO does not work with inter-base-station downlink joint transmission.
Downlink channel prediction for mobility enhancement	DL_SU_PREDICT_MOBILITY_ENH_SW option of the NRDUCellAlgoSwitch.SuMimoMultipleLayerSw parameter	<i>MIMO (TDD)</i>	Distributed massive MIMO does not work with downlink channel prediction for mobility enhancement.
Intelligent uplink multi-layer SINR prediction	UL_SU_SINR_INTEL_PREDICT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter	<i>MIMO (TDD)</i>	Distributed massive MIMO does not work with intelligent uplink multi-layer SINR prediction.
Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with inter-cell interference avoidance.

Function Name	Function Switch	Reference	Description
LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	Distributed massive MIMO does not work with LTE TDD and NR spectrum sharing.
SCell CSI-RS beam sweeping	CARR_CSI_BEAM_SWEEPING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>Carrier Aggregation</i>	Distributed massive MIMO does not work with SCell CSI-RS beam sweeping.
Rank selection algorithm	SRS_DIFF_BASED_RANK_SELECT_SW option of the NRDUCellAlgoSwitch.RankSelectAlgoSw parameter	None	Distributed massive MIMO does not work with the rank selection algorithm.
Downlink mobile MU-MIMO enhancement	DL_MOBILE_MU_ENH_SW option of the NRDUCellAlgoSwitch.UeFeatureBasedPairEnhSw parameter	None	Distributed massive MIMO does not work with downlink mobile MU-MIMO enhancement.
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with Co-MM.
VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	Distributed massive MIMO does not work with VMIMO in DAS-based indoor coverage.
Uplink low-PAPR DMRS	UL_LOW_PAPR_DMRS_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with uplink low-PAPR DMRS.
Time-domain adaptive filtering	PUSCH_TIME_DOM_ADAPT_FILTER_SW option of the NRDUCellChnCovAlgo.UICoverageAlgoSwitch parameter	<i>Turbo Uplink (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with time-domain adaptive filtering.
CSI-RS beam sweeping	CSIRS_BEAM_SWEEPING_SW option of the NRDUCellCsirs.CsiSwitch parameter	None	Distributed massive MIMO does not work with CSI-RS beam sweeping.
IAB	NRDUCell.DeployPosition set to DEPLOY_IAB	None	Distributed massive MIMO does not work with IAB.

Function Name	Function Switch	Reference	Description
Multi-RF-module coordinated energy saving	ENERGY_EFFICIENCY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter ENERGY_EFF_BASED_COLLAB_SCH_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter PWR_EFF_MODEL_BUILD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter PWR_EFF_MODEL_INTEL_BUILD_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If distributed massive MIMO is enabled, the following functions cannot be enabled: normal building of module power consumption models, intelligent building of module power consumption models, energy-efficiency-based inter-frequency handover, and energy-efficiency-based coordinated scheduling.
TTI-level channel shutdown	TTI_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter NRDUCellPowerSaving.ComChnDynShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with TTI-level channel shutdown.
Intelligent power control energy saving	SMART_PWR_CTRL_ENERGY_SAVE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with intelligent power control energy saving.
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Distributed massive MIMO does not work with High-speed Railway Superior Experience.
Proactive avoidance of remote interference	SRS_INTRF_AVOID_PUSCH_SCH_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with proactive avoidance of remote interference.
Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with remote interference adaptive avoidance.

Function Name	Function Switch	Reference	Description
Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	Distributed massive MIMO does not work with enhanced remote interference adaptive avoidance.
Low-power RF channel dynamic shutdown	LOW_POWER_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with low-power RF channel dynamic shutdown.
Dynamic adjustment of the RMSI broadcast interval	NRDUCellPowerSaving.RmsiAdjustMode set to BEAM	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with dynamic adjustment of the RMSI broadcast interval.
Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with uplink symbol power saving.
pRRU-level precise symbol power saving	DL_PRECISE_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitchExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with pRRU-level precise symbol power saving.
Dynamic RF channel shutdown	RF_CHN_DYN_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitchExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with dynamic RF channel shutdown.
Common-channel dynamic shutdown	NRDUCellPowerSaving.ComChnDynShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with common-channel dynamic shutdown.
Multi-carrier joint carrier shutdown	MULTI_CARR_JOINT_CARR_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitchExtSwitch parameter	<i>Multi-RAT Coordinated Energy Saving and Energy Conservation and Emission Reduction</i>	Distributed massive MIMO does not work with multi-carrier joint carrier shutdown.

Function Name	Function Switch	Reference	Description
LNR coordinated channel shutdown	MULTI_RAT_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	Distributed massive MIMO does not work with LNR coordinated channel shutdown.
NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Distributed massive MIMO does not work with NR inter-carrier dynamic power sharing.
LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW and LTE_NR_POWER_SHARING_OPT_SW options of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	Distributed massive MIMO does not work with LTE TDD and NR Flash Dynamic Power Sharing.
NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	Distributed massive MIMO does not work with NR multi-band power adaptive scheduling.
Full-buffer UE-specific CSI-RS for CM	NRDUCellCsirs.CsiUserPeriod	<i>Channel Management</i>	This parameter must be set to NOT_CONFIG for distributed massive MIMO cells.
3D coverage pattern	NRDUCellTrpBeam.CoverageScenario	<i>Beam Management</i>	This parameter must be set to DEFAULT for distributed massive MIMO cells.
Extended Cell Range of 90 km (TDD)	NRDUCell.CellRadius	<i>Extended Cell Range</i>	This parameter must be set to a value greater than 60000 and less than or equal to 90000 distributed massive MIMO cells.

- Enhanced functions of distributed massive MIMO

Function Name	Function Switch	Reference	Description
Downlink beam selection for distributed massive MIMO	DM_MIMO_DL_BEAM_SELECT_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	TRP-based weight selection for distributed massive MIMO does not work with downlink beam selection for distributed massive MIMO.
Frequency-domain high-resolution power control	FREQ_HIGH_RESOLUT_PWR_CTL_SW option of the NRDUCellPdsch.DlPdschAlgoSwitch parameter	<i>MIMO (TDD)</i>	TRP-based weight selection for distributed massive MIMO does not work with frequency-domain high-resolution power control.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	MU-MIMO	NRDUCellAlgoSwitch.MuMimoSwitch	<i>MIMO (TDD)</i>	For UEs not in overlapping areas, the beams of multiple TRPs can use the same time-frequency resources of PUSCH, PDSCH, or PDCCH for spatial multiplexing if the UL_MU_MIMO_SW , DL_MU_MIMO_SW , or PDCCH_MU_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected.
TDD	PUCCH structure type	NRDUCellPucch.StructureType	<i>Channel Management</i>	This parameter must be set to LONG_STRUCTURE for distributed massive MIMO cells.
TDD	CSI-RS resource	NRDUCellCsirs.CsirsCellResourceNum	<i>Channel Management</i>	A distributed massive MIMO cell supports only one CSI-RS resource (with the value 1_RESOURCE always taking effect for the NRDUCellCsirs.CsirsCellResourceNum parameter).

RAT	Function Name	Function Switch	Reference	Description
TDD	Broadcast beam densification	SCENARIO_BEAM_DENSIFY_ALGO_SW option of the NRDUCellTrpBeam.ScenarioBeamAlgoSw parameter	<i>Beam Management</i>	If the maximum number of SSB beams supported under the current slot configuration is required in a distributed massive MIMO cell, broadcast beam densification can be enabled.
TDD	PRACH fixed delay compensation	NRDUCellTrpAlgo.FixedDelayCompensation	<i>Random Access Control</i>	For multiple TRPs of a distributed massive MIMO cell, the fixed delay compensation function takes effect only based on the fixed delay configured for the master TRP.
TDD	Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutofsyncDetSwitch parameter	<i>Synchronization</i>	Clock out-of-synchronization detection does not take effect in distributed massive MIMO cells.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For LampSite base stations:

All NR TDD-capable main control boards support this function. For details, see *LampSite BBU Hardware Description* in *3900 & 5900 Series Base Station Product Documentation*.

For macro base stations:

All NR TDD-capable main control boards support this function. For baseband processing units, only the UBBPg2a, UBBPg3, UBBPg3b, UBBPh3a, and UBBPh3b support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

 NOTE

For LampSite base stations:

One BBP supports only one distributed massive MIMO cell. It does not support the mixed configuration of distributed and non-distributed massive MIMO cells.

For macro base stations:

In TDD, one BBP supports only one distributed massive MIMO cell. It does not support the mixed configuration of distributed and non-distributed massive MIMO cells.

RF Modules

For LampSite base stations:

- All NR TDD-capable 4T4R pRRUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- All RHUBs, except the RHUB592x, RHUB593x, RHUB595x, and RHUB39xx, support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

For macro base stations:

- Only RF modules of the same type can be combined to serve a distributed massive MIMO cell.
- The difference in the lengths of CPRI optical fibers connected to different RF modules cannot exceed 2 km.
- In TDD, the following RF modules support this function: 8T8R-capable AAU5243 in ToB scenarios.

4.3.4 Cells

The **NRDUCell.DuplexMode** parameter must be set to **TDD**, and the **NRDUCellTrp.TxRxMode** parameter must be set to **8T8R**, **4T4R**, or **2T2R**.

Distributed massive MIMO requires that the **NRDUCell.CellRadius** parameter be set to a value less than or equal to **14500**.

4.3.5 Others

- For LampSite base stations, 4T4R TRPs or both 4T4R and 2T2R TRPs can be combined to serve one distributed massive MIMO cell and these TRPs must be configured on BBPs in the same BBU. In addition, pRRUs in different TX/RX modes cannot be used together. Each distributed massive MIMO cell can be served by a maximum of 96 pRRUs.
- For TDD macro base stations, either 4T4R or 8T8R RF modules can be combined to serve one distributed massive MIMO cell; RF modules in different TX/RX modes cannot be used together. In single-carrier networking with 4T4R TRPs, a maximum of 12 TRPs can be combined to serve one distributed massive MIMO cell. In single-carrier networking with 8T8R TRPs, a maximum of 6 TRPs can be combined to serve one distributed massive MIMO cell. In addition, these TRPs must be configured on BBPs in the same BBU.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment. For details about these parameters, see *Cell Management*.

Table 4-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
TRP Type	NRDUCellTrp.TrpType	Set this parameter to MASTER_DM_MIMO for a master TRP or SLAVE for a slave TRP.
Master TRP ID	NRDUCellTrp.MasterTrpId	For a slave TRP, set this parameter to the value of NRDUCellTrp.NrDuCellTrpId of the master TRP, associating the slave TRP with the master TRP.
Distributed MIMO Switch	NRDUCellAlgoSwitch.DmMimoSwitch	Select the DM_MIMO_SERVICE_SWITCH option.
Distributed MIMO Switch	NRDUCellAlgoSwitch.DmMimoSwitch	It is recommended that the DM_MIMO_TRP_SEL_WEIGHT_SW option be selected in light- or medium-load scenarios.
Distributed MIMO Switch	NRDUCellAlgoSwitch.DmMimoSwitch	It is recommended that the DM_MIMO_PDCCH_BEAM_OPTION_SWITCH option be selected in heavy-load scenarios where PDCCH resources are insufficient.

Parameter Name	Parameter ID	Setting Notes
Distributed MIMO Switch	NRDUCellAlgoSwitch.DmMimoSwitch	It is recommended that the DM_MIMO_PDCCH_TRP_TRANS_SW option be selected for a heavy-load LampSite cell that experiences severe interference from neighboring macro cells and has insufficient PDCCH resources.
Distributed MIMO Switch	NRDUCellAlgoSwitch.DmMimoSwitch	It is recommended that the DM_MIMO_UL_BEAM_BOOST_SWITCH option be selected for cells served by 8T8R TRPs in ToB scenarios.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

For configuration examples of distributed massive MIMO cells, see "Establishing a TDD Cell" and "Establishing an RF Combination Cell" in the "Using MML Commands" section in *Cell Management*. The following provides only the differences in function activation.

- Configuring TRPs for a distributed massive MIMO cell
//Configuring the TRP type and binding TRPs
MOD NRDUCELLTRP: NrDuCellTrpId=1, NrDuCellId=0, TxRxMode=4T4R,
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=200,
CpriCompression=3DOT2_COMPRESSION, TrpType=MASTER_DM_MIMO;
MOD NRDUCELLTRP: NrDuCellTrpId=2, NrDuCellId=65535, TxRxMode=4T4R, TrpType=SLAVE,
MasterTrpId=1;
- Enabling distributed massive MIMO
//Enabling distributed massive MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_SERVICE_SWITCH-1;
//Enabling PDCCH beam optimization for distributed massive MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
DmMimoSwitch=DM_MIMO_PDCCH_BEAM_OPT_SWITCH-1;
//Enabling TRP-level PDCCH transmission for distributed massive MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_PDCCH_TRP_TRANS_SW-1;
//Enabling TRP-based weight selection for distributed massive MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_TRP_SEL_WEIGHT_SW-1;
//Enabling uplink beam boosting for distributed massive MIMO in ToB scenarios

```
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,  
DmMimoSwitch=DM_MIMO_UL_BEAM_BOOST_SWITCH-1;
```

Deactivation Command Examples

Before deactivating distributed massive MIMO, deactivate the cell. To associate a slave TRP with the deactivated DU cell, add the TRP only after removing the original DU cell TRP. For details about how to establish a DU cell, see "Establishing a TDD Cell" and "Establishing an RF Combination Cell" in the "Using MML Commands" section in *Cell Management*. The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling TRP-level PDCCH transmission for distributed massive MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_PDCCH_TRP_TRANS_SW-0;  
//Disabling PDCCH beam optimization for distributed massive MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_PDCCH_BEAM_OPT_SWITCH-0;  
//Disabling TRP-based weight selection for distributed massive MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_TRP_SEL_WEIGHT_SW-0;  
//Disabling uplink beam boosting for distributed massive MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_UL_BEAM_BOOST_SWITCH-0;  
//Deactivating the cell  
DEA NRCELL: NrCellId=0;  
//Disabling distributed massive MIMO  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, DmMimoSwitch=DM_MIMO_SERVICE_SWITCH-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Run the **DSP NRDUCELL** command to check the status of a distributed massive MIMO cell. If the value of **NR DU Cell State** is **Normal**, the cell has been successfully set up.

Run the **DSP NRDUCELLTRP** command to check the status of TRPs. If the value of **NR DU Cell TRP State** is **Available**, the TRPs have been successfully set up.

The results of interference monitoring under cell performance monitoring on the MAE are not applicable to distributed massive cells. The average interference of antennas cannot reflect the actual interference of the distributed massive MIMO cell as the interference of each antenna varies greatly with the antenna location.

4.4.3 Network Monitoring

For LampSite base stations:

- Monitor the increase in UE throughput by comparing the following indicators of a distributed massive MIMO cell and those of the original cells.
 - **User Uplink Average Throughput (DU)**
 - **User Downlink Average Throughput (DU)**

- Monitor the increase in the average cell capacity by comparing the following indicators of a distributed massive MIMO cell and those of the original cells.
 - **Cell Uplink Average Throughput (DU)**
 - **Cell Downlink Average Throughput (DU)**

For macro base stations:

- Monitor the increase in UE throughput by comparing the following indicators of the distributed massive MIMO cell and those of the original cells.
 - Average downlink UE throughput (Mbit/s) = $(\text{N.ThpVol.DL} - \text{N.ThpVol.DL.LastSlot}) / \text{N.ThpTime.DL.RmvLastSlot} \times 1000$. [Table 4-7](#) lists the related performance counters.

Table 4-7 Related performance counters

Counter ID	Counter Name
1911816643	N.ThpVol.DL
1911816848	N.ThpVol.DL.LastSlot
1911816850	N.ThpTime.DL.RmvLastSlot

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Energy Conservation and Emission Reduction*
- *UE Power Saving*
- *Downlink Scheduling*
- *Channel Management*
- *Random Access Control*
- *MIMO (TDD)*
- *Beam Management*
- *CoMP*
- *Co-MM (Low-Frequency TDD)*
- *High Speed Mobility*
- *Multi-Operator Sharing*
- *Hyper Cell*
- *Remote Interference Management (Low-Frequency TDD)*
- *Indoor Coverage*
- *Cell Combination*
- *Cell Management*
- *Carrier Aggregation*
- *Super Uplink*
- *Power Control*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Multi-RAT Coordinated Energy Saving*
- *NR Inter-Carrier Dynamic Power Allocation*
- *LTE and NR Power Allocation*
- *License Control Item Lists*
- *Extended Cell Range*
- *Synchronization*

- *Turbo Uplink (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*